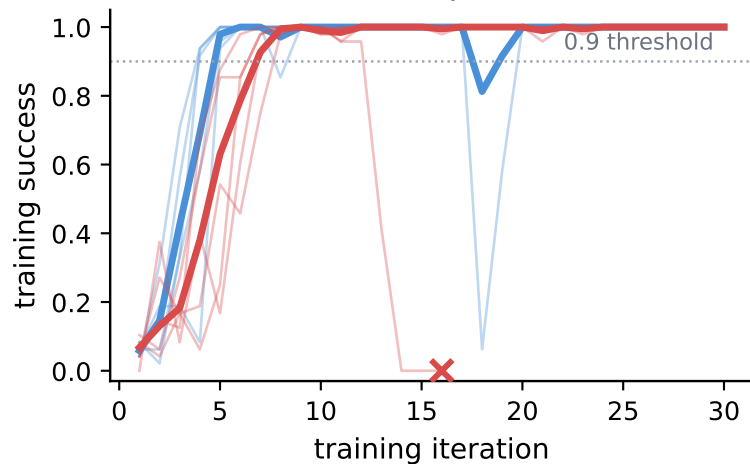
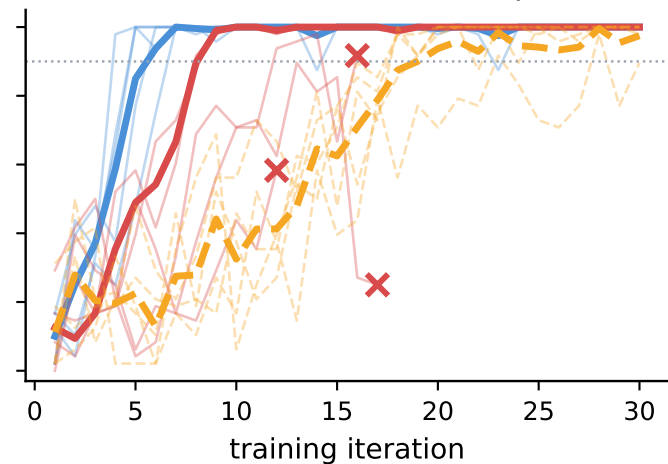


(a) Qwen3-4B, matched optimizer (5 seeds each)



(b) Qwen3-30B-A3B (5 seeds per arm)



— aligned potential (Muon), 5/5 stable — outcome-only (Muon), 1/5 diverge — aligned potential (Muon), 5/5 stable — outcome-only (AdamW), 5/5 stable
— outcome-only (Muon), 3/5 diverge